

Foundation Language Models vs. Fine-Tuned Language Models

Foundation language models, such as the aforementioned MT-NLG and GPT-3, are what is usually referred to when discussing LLMs. They're trained on vast amounts of data and can perform a wide variety of NLP tasks, from answering questions and generating book summaries to completing and translating sentences.

Thanks to their size, foundation models can perform well even when they have little domain-specific data at their disposal. They have good general performance across tasks but may not excel at performing any one specific task.

Fine-tuned language models, on the other hand, are large language models derived from foundation LLMs. They're customized for specific use cases or domains and, thus, become better at performing more specialized tasks.

Apart from the fact that fine-tuned models can perform specific tasks better than foundation models, their biggest strength is that they are lighter and, generally, easier to train. But how does one actually fine-tune a foundation model for specific objectives?

Currently, the most popular method is customizing a model using parameter-efficient customization techniques, such as p-tuning, prompt tuning, adapters, and so on. Customization is far less time-consuming and expensive than fine-tuning the entire model, although it may lead to somewhat poorer performance than other methods. Customization methods are further discussed in [Part 3](#).

Evolution of Large Language Models

AI systems were historically about processing and analyzing data, not generating it. They were more oriented toward perceiving and understanding the world around us rather than on generating new information. This distinction marks the main difference between *Perceptive AI* and *Generative AI*, with the latter becoming increasingly prevalent since around 2020, or after companies started adopting transformer models and developing increasingly more robust LLMs at a large scale.

The advent of large language models further fueled a revolutionary paradigm shift in the way NLP models are designed, trained, and used. To truly understand this, it may be helpful to compare large language models to previous NLP models and how they worked. For this purpose, let's briefly explore three regimes in the history of NLP: pre-transformers NLP, transformers NLP, and LLM NLP.

1. **Pre-transformers NLP** was mainly marked by models that relied on human-crafted rules rather than machine learning algorithms to perform NLP tasks. This made them suitable for simpler tasks that didn't require too many rules, like text classification, but unsuitable for more complex tasks, such as machine translation. Rule-based models also performed poorly in edge-case scenarios because they couldn't make accurate predictions or classifications for never-before-seen data for which no clear rules were set. This problem was somewhat solved with simple neural networks, such as RNNs and LSTMs, developed during the later phases of this period. RNNs and LSTMs could memorize past data to a certain extent and, thus, provide context-dependent predictions and

classifications. However, RNNs and LSTMs could not make predictions over long spans of text, limiting their effectiveness.

2. **Transformers NLP** was set in motion by the rise of the transformer architecture in 2017. Transformers could generalize better than the then-prevailing RNNs and LSTMs, capture more context, and process more data at once. These improvements enabled NLP models to understand longer sequences of data and perform a much wider range of tasks. However, from today's point of view, models developed during this period had limited capabilities, mainly due to the general lack of large-scale datasets and adequate computational resources. They also mainly sparked attention among researchers and experts in the field but not the general public, as they weren't user-friendly nor accurate enough to become commercialized.
3. **LLM NLP** was mainly initiated by the launch of OpenAI's GPT-3 in 2020. Large language models like GPT-3 were trained on massive amounts of data, which allowed them to produce more accurate and comprehensive NLP responses compared to previous models. This unlocked many new possibilities and brought us closer to achieving what many consider "true" AI. Also, LLMs made NLP models much more accessible to non-technical users who could now solve a variety of NLP tasks just by using natural-language prompts. *NLP technology was finally democratized.*

The switch from one methodology to another was largely driven by relevant technological and methodological advancements, such as the advent of neural networks, attention mechanisms, and transformers and developments in the field of unsupervised and self-supervised learning. The following sections will briefly explain these concepts, as understanding them is crucial for truly understanding how LLMs work and how to build new LLMs from scratch.

Neural Networks

Neural networks (NNs) are machine learning algorithms loosely modeled after the human brain. Like the biological human brain, artificial neural networks consist of neurons, also called nodes, that are responsible for all model functions, from processing input to generating output.

The neurons are further organized into layers, vertically stacked components of NNs that perform specific tasks related to input and output sequences.